

# Adaptive Heuristic Portfolio TSP Report

## Overview

- Condition count: 8.
- Best held-out TSPLIB gap: phase8\_best\_single\_fixed\_heuristic.
- Best combined transfer gap: phase8\_oracle\_selector.
- Lowest selector regret: phase8\_best\_single\_fixed\_heuristic.
- Pareto-efficient conditions: phase8\_best\_single\_fixed\_heuristic, phase8\_full\_solver\_evolution, phase8\_llm\_evolved\_controller\_diversity\_failure\_replay.

## Run Metadata

- run\_name: run\_20260520\_222400\_o
- started\_at\_local: 2026-05-20 22:24:00
- finished\_at\_local: 2026-05-20 22:42:52
- duration\_hhmm: 00:19
- duration\_seconds: 1131.434
- seed\_offset: 14000
- replicate\_label: o
- judge\_status: enabled
- judge\_provider: openai
- judge\_model: gpt-4.1-mini

## Condition Comparison

| Condition                          | Mode             | Replay | Final TSPLIB Gap | Selector Regret | Run time (ms) | Runtime -Adjusted Gap | Family Gap | Transfer Gap | Mean Novelty | Mean Complexity | Pareto Efficient |
|------------------------------------|------------------|--------|------------------|-----------------|---------------|-----------------------|------------|--------------|--------------|-----------------|------------------|
| phase8_best_single_fixed_heuristic | best_fixed       | none   | 0.07226          | 0.0             | 702.170743    | 0.07226               | 0.007284   | 0.03122      | 0.0          | 0.0             | True             |
| phase8_random_portfolio            | random_portfolio | none   | 0.19294          | 0.12068         | 1752.028029   | 0.481416              | 0.074127   | 0.1179       | 0.0          | 0.0             | False            |
| phase8_oracle_selector             | oracle_selector  | none   | 0.07226          | 0.0             | 704.727914    | 0.072523              | 8.9e-05    | 0.026678     | 0.0          | 0.0             | False            |

| Condition                                              | Mode                | Replay            | Final TSPLI B Gap | Selector Regret | Runtime (ms) | Runtime -Adjusted Gap | Family Gap | Transfer Gap | Mean Novelty | Mean Complexity | Pareto Efficient |
|--------------------------------------------------------|---------------------|-------------------|-------------------|-----------------|--------------|-----------------------|------------|--------------|--------------|-----------------|------------------|
| phase8_supervised_ml_selector                          | supervised_selector | none              | 0.07226           | 0.0             | 716.268414   | 0.073711              | 0.007284   | 0.03122      | 0.0          | 0.0             | False            |
| phase8_llm_static_selector                             | static_selector     | none              | 0.120261          | 0.048001        | 3584.457971  | 0.613911              | 0.001857   | 0.045479     | 0.0          | 0.62            | False            |
| phase8_llm_evolved_adaptive_controller                 | adaptive_controller | none              | 0.120261          | 0.048001        | 5351.889486  | 0.91662               | 0.0090132  | 0.101232     | 0.076586     | 0.62            | False            |
| phase8_llm_evolved_controller_diversity_failure_replay | adaptive_controller | diversity_failure | 0.201598          | 0.129338        | 137.508014   | 0.201598              | 0.045212   | 0.102828     | 0.0          | 0.62            | True             |
| phase8_full_solver_evolution                           | full_solver         | none              | 0.213387          | 0.141127        | 101.469429   | 0.213387              | 0.150269   | 0.173523     | 0.849625     | 0.56            | True             |

## Condition Notes

### phase8\_best\_single\_fixed\_heuristic

- Execution mode best\_fixed with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 702.170743 ms, runtime inflation 0.0, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.
- Fixed heuristic choice: farthest\_insertion\_two\_opt.

### phase8\_random\_portfolio

- Execution mode random\_portfolio with replay none.
- Final held-out gap 0.19294, selector regret 0.12068, runtime-adjusted gap 0.481416.
- Family transfer 0.074127 and combined transfer 0.1179.
- Runtime 1752.028029 ms, runtime inflation 1.49516, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

### phase8\_oracle\_selector

- Execution mode oracle\_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.072523.

- Family transfer 8.9e-05 and combined transfer 0.026678.
- Runtime 704.727914 ms, runtime inflation 0.003642, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

### **phase8\_supervised\_ml\_selector**

- Execution mode supervised\_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.073711.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 716.268414 ms, runtime inflation 0.020077, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

### **phase8\_llm\_static\_selector**

- Execution mode static\_selector with replay none.
- Final held-out gap 0.120261, selector regret 0.048001, runtime-adjusted gap 0.613911.
- Family transfer 0.001857 and combined transfer 0.045479.
- Runtime 3584.457971 ms, runtime inflation 4.104824, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: annealed\_multi\_start:cheapest\_insertion\_two\_opt:farthest\_insertion\_two\_opt:edge\_preserving\_restart:stag3:cand-2:restart1.
- Controller rule summary: {"acceptance\_bias": 0.006, "candidate\_limit\_offset": -2, "clustered\_heuristic": "cheapest\_insertion\_two\_opt", "corridor\_heuristic": "limited\_three\_opt", "default\_heuristic": "farthest\_insertion\_two\_opt", "description": "Deterministic instance-adaptive selector over frozen TSP heuristic portfolio; chooses structure-appropriate construction/refinement with conservative online tuning knobs.", "failed\_perturbation\_threshold": 2, "grid\_like\_heuristic": "candidate\_pruned\_two\_opt", "low\_improvement\_threshold": 0.08, "name": "deterministic\_instance\_adaptive\_oracleish\_baseline", "nearest\_neighbor\_trap\_heuristic": "cheapest\_insertion\_two\_opt", "random\_like\_heuristic": "annealed\_multi\_start", "restart\_offset": 1, "stagnation\_switch\_heuristic": "edge\_preserving\_restart", "stagnation\_threshold": 3, "temperature\_scale": 1.15, "time\_budget\_trigger": 0.55, "two\_cluster\_bottleneck\_heuristic": "farthest\_insertion\_two\_opt"}

### **phase8\_llm\_evolved\_adaptive\_controller**

- Execution mode adaptive\_controller with replay none.
- Final held-out gap 0.120261, selector regret 0.048001, runtime-adjusted gap 0.91662.
- Family transfer 0.090132 and combined transfer 0.101232.
- Runtime 5351.889486 ms, runtime inflation 6.62192, mean novelty 0.076586, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: nearest\_neighbor\_multistart:cluster\_first\_local\_search:edge\_preserving\_restart:annealed\_multi\_start:stag3:cand-1:restart2.
- Controller rule summary: {"acceptance\_bias": 0.007, "candidate\_limit\_offset": -1, "clustered\_heuristic": "cluster\_first\_local\_search", "corridor\_heuristic": "limited\_three\_opt", "default\_heuristic": "farthest\_insertion\_two\_opt", "description": "Deterministic instance-adaptive selector/scheduler over frozen TSP heuristics with structure-triggered switching and bounded parameter offsets.", "failed\_perturbation\_threshold": 3, "grid\_like\_heuristic": "candidate\_pruned\_two\_opt", "low\_improvement\_threshold": 0.12, "name": "interp\_adaptive\_tsp\_hh\_v2", "nearest\_neighbor\_trap\_heuristic": "cheapest\_insertion\_two\_opt",

"random\_like\_heuristic": "nearest\_neighbor\_multistart", "restart\_offset": 2, "stagnation\_switch\_heuristic": "annealed\_multi\_start", "stagnation\_threshold": 3, "temperature\_scale": 1.12, "time\_budget\_trigger": 0.6, "two\_cluster\_bottleneck\_heuristic": "edge\_preserving\_restart"}

### phase8\_llm\_evolved\_controller\_diversity\_failure\_replay

- Execution mode adaptive\_controller with replay diversity\_failure.
- Final held-out gap 0.201598, selector regret 0.129338, runtime-adjusted gap 0.201598.
- Family transfer 0.045212 and combined transfer 0.102828.
- Runtime 137.508014 ms, runtime inflation -0.804167, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: True.
- Controller signature: nearest\_neighbor\_multistart:cluster\_first\_local\_search:edge\_preserving\_restart:annealed\_multi\_start:stag3:cand2:restart1.
- Controller rule summary: {"acceptance\_bias": 0.006, "candidate\_limit\_offset": 2, "clustered\_heuristic": "cluster\_first\_local\_search", "corridor\_heuristic": "limited\_three\_opt", "default\_heuristic": "farthest\_insertion\_two\_opt", "description": "Deterministic instance-adaptive controller selecting from frozen TSP heuristic portfolio using structure descriptors and online stagnation signals to schedule bounded escape/exploit behavior.", "failed\_perturbation\_threshold": 2, "grid\_like\_heuristic": "candidate\_pruned\_two\_opt", "low\_improvement\_threshold": 0.08, "name": "interpretable\_instance\_adaptive\_hh\_v1", "nearest\_neighbor\_trap\_heuristic": "candidate\_pruned\_two\_opt", "random\_like\_heuristic": "nearest\_neighbor\_multistart", "restart\_offset": 1, "stagnation\_switch\_heuristic": "annealed\_multi\_start", "stagnation\_threshold": 3, "temperature\_scale": 1.15, "time\_budget\_trigger": 0.6, "two\_cluster\_bottleneck\_heuristic": "edge\_preserving\_restart"}

### phase8\_full\_solver\_evolution

- Execution mode full\_solver with replay none.
- Final held-out gap 0.213387, selector regret 0.141127, runtime-adjusted gap 0.213387.
- Family transfer 0.150269 and combined transfer 0.173523.
- Runtime 101.469429 ms, runtime inflation -0.855492, mean novelty 0.849625, and complexity 0.56.
- Pareto efficient: True.

## Judge Appendix

### Conservative Interpretation of TSP Adaptive Heuristic Portfolio Results

| Condition                              | Held-out TSPLIB gap | Selector regret vs oracle | Runtime-adjusted gap | Runtime inflation | Transfer gap | Pareto efficient | Comments                                                                                                                                         |
|----------------------------------------|---------------------|---------------------------|----------------------|-------------------|--------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| **phase8_best_single_fixed_heuristic** | **0.07226**         | 0.0                       | 0.07226              | 0.0               | 0.031222     | Yes              | Best baseline on held-out TSPLIB. Simple fixed heuristic (farthest_insertion_two_opt) with zero added runtime or regret. Strong static baseline. |
| **phase8_oracle_selector**             | 0.07226             | 0.0                       | 0.07252              | 0.0036            | 0.026678     | No               | Oracle selector is an upper bound, matches best_fixed in TSPLIB gap, negligible runtime inflation. Not deployable but useful reference.          |

| Condition                                                         | Held-out TSPLIB gap | Selector regret vs oracle | Runtime-adjusted gap | Runtime inflation | Transfer gap | Pareto efficient | Comments                                                                                                                                                    |
|-------------------------------------------------------------------|---------------------|---------------------------|----------------------|-------------------|--------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>**phase8_supervised_ml_selector**</b>                          | 0.07226             | 0.0                       | 0.07371              | 0.0201            | 0.031222     | No               | Matches best TSPLIB gap, minor selector regret and slight runtime inflation.                                                                                |
| <b>**phase8_llm_static_selector**</b>                             | 0.12026             | 0.048                     | 0.61391              | 4.10              | 0.04548      | No               | Worse held-out gap and large runtime inflation (~4x). Higher regret. Despite instance-adaptive mechanism, runtime cost is high and gap is larger.           |
| <b>**phase8_llm_evolved_adaptive_controller**</b>                 | 0.12026             | 0.048                     | 0.91662              | 6.62              | 0.101232     | No               | Adaptive controller has highest runtime inflation and worst runtime adjusted gap. Selector regret moderate; held-out gaps not improved over fixed baseline. |
| <b>**phase8_llm_evolved_controller_diversity_failure_replay**</b> | 0.20160             | 0.1293                    | 0.20160              | -0.80             | 0.102828     | Yes              | Lower runtime (~-0.8 runtime inflation) but much worse held-out gap and high selector regret vs oracle. Still pareto because of low runtime.                |
| <b>**phase8_random_portfolio**</b>                                | 0.19294             | 0.121                     | 0.48142              | 1.50              | 0.11790      | No               | Poor held-out gap and high regret relative to oracle. Significant runtime inflation; not pareto.                                                            |
| <b>**phase8_full_solver_evolution**</b>                           | 0.21339             | 0.141                     | 0.21339              | -0.86             | 0.17352      | Yes              | Worse held-out gap and regret; lowest runtime (negative runtime inflation). Pareto due to runtime saving but less effective on TSPLIB gaps.                 |

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## Summary & Prioritized Conclusions

### - **\*\*Primary endpoint (held-out TSPLIB gap):\*\***

The **\*\*best performer\*\*** is the **\*phase8\_best\_single\_fixed\_heuristic\*** with gap 0.07226, matching the oracle selector's gap. This shows that the simple fixed heuristic **farthest\_insertion\_two\_opt** is difficult to beat on this metric.

### - **\*\*Selector regret vs oracle:\*\***

The oracle selector sets a zero regret baseline. The **\*supervised ML selector\*** matches the oracle's held-out gap without selector regret and only minor runtime inflation, suggesting effective instance-adaptive control.

### - **\*\*Runtime-adjusted gap & inflation:\*\***

The fixed heuristic and oracle selector show the best tradeoff, with low runtime and gaps. Adaptive controllers and static LLM-based selectors inflate runtime substantially (up to ~6.6x), harming their practical deployability.

### - **\*\*Pareto efficiency:\*\***

Pareto-optimal conditions include the fixed heuristic, a low-runtime evolved controller with diversity failure replay, and a full solver evolution with low runtime cost. However, none improve upon or match the fixed heuristic's held-out gap with reasonable runtime.

- **Cross-family transfer:**

Transfer gaps are lowest for the fixed heuristic and oracle selector (~0.03), increasing for adaptive controllers (~0.10+), indicating less generalizable control learned by evolution or LLM approaches.

- **Interpretability & adaptive control claims:**

While adaptive controllers show some interpretable structure-triggered switching, their runtime inflation and worse performance vs fixed suggest limited practical benefit at this phase. The supervised ML selector approximates oracle performance without excessive runtime increase, offering a more interpretable and deployable instance-adaptive control baseline, rather than discovering new heuristics.

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## Recommendations

- Emphasize the strong baseline of the fixed heuristic and supervised ML selector in matching oracle-level gaps with manageable runtime, highlighting stable and interpretable instance-adaptive control.
- Be cautious claiming benefits of LLM static or evolved adaptive controllers: they incur excessive runtime inflation and do not improve held-out gaps or regret substantially.
- Use oracle selector as a performance upper bound only, not a deployable baseline.
- Future work should focus on reducing runtime inflation in adaptive controllers while maintaining or improving held-out gap and selector regret.

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**In brief:**

> The fixed heuristic *farthest\_insertion\_two\_opt* remains the best held-out TSPLIB solution baseline, closely matched by the supervised ML selector with negligible selector regret and small runtime inflation. Oracle selector establishes a gap lower bound but is impractical. Adaptive controllers and LLM-driven selectors introduce excessive runtime inflation without gap improvement, limiting their practical instance-adaptive utility in this phase.